COMPUTER NETWORKS LAB | 23CS2005

Cisco Packet Tracer Instruction Guide

Experiments 2 to 9 | Built from the lab record: topologies, addressing tables, configuration commands, verification and viva questions

Contents

Exp	Topic (as in the record)	Key skill
2	Subnetting	/25 subnetting of 192.168.10.0/24, router gateways
3	Simulation of LAN Topology with DNS and DHCP	DNS, DHCP and HTTP servers, two LANs
4	Static Routing (two LANs over a serial WAN)	ip route, serial DCE link, clock rate
5	Dynamic Routing using OSPF (three routers)	Single-area OSPF over a serial triangle
6	Intra-VLAN and Inter-VLAN Routing	Access ports, trunk, router-on-a-stick
7	IPv6 Addressing using SLAAC	ipv6 unicast-routing, EUI-64, RA
8	Static NAT, Dynamic NAT and PAT	Inside/outside, pool, overload
9	Bus, Star, Ring and Mesh topologies	Switch-only topologies, STP behaviour

Also included: **Section A** (Packet Tracer basics and cabling), **Section B** (troubleshooting and command cheat sheet) and **Section C** (points in the current record that should be corrected before submission).

Every IP address, device name and interface in this guide comes from the lab record. Where the record contradicts itself, the guide uses the version that works and Section C explains the change.

Section A: Packet Tracer basics

A.1 Workspace in 60 seconds

- **Device palette** (bottom-left): choose a category (Network Devices, End Devices, Connections), then a sub-category, then drag the model in.
- **Realtime vs Simulation** (bottom-right): Simulation mode lets you step through each packet. Use it for the "Simulation Mode" screenshots in your record.
- **Add Simple PDU** (closed envelope icon): click source then destination for a one-click ping test; the Last Status column shows Successful or Failed.

A.2 Which cable to use

Connection	Cable	Notes
PC, server or laptop to switch; switch to router	Copper Straight-Through	Solid line
Router to router (Ethernet), PC to router, switch to switch	Copper Cross-Over	Dashed line
Router to router over serial (Ex 4, 5, 8)	Serial DCE	Red line; the router at the clock icon end needs `clock rate 64000`
Not sure	Automatically Choose Connection Type	Lightning bolt icon

A link is up when both end dots are **green**. Orange dots on a switch mean it is still in spanning-tree listening/learning, so wait about 30 seconds (or use the fast-forward button).

A.3 Interface names depend on the router model

The record uses the generic **Router-PT** (interfaces `FastEthernet0/0`, `FastEthernet1/0`, `Serial2/0`, `Serial3/0`) in some experiments and the **2911** (`GigabitEthernet0/0` and `0/1`) in others. Commands in this guide use the names shown in the topology of each experiment. If your port is called something different, run `show ip interface brief` and substitute the names you see.

A.4 Configuring end devices and routers

- PC, laptop or server: click it, **Desktop, IP Configuration**; choose Static or DHCP and fill address, mask, gateway, DNS. **Desktop, Command Prompt** gives ping, ipconfig, tracert and nslookup.
- Server services (DNS, DHCP, HTTP): **Services** tab.
- Router or switch: **CLI** tab. Answer **no** to the initial configuration dialog.

Prompt	Mode	Enter with
`Router>`	User EXEC	`default`
`Router#`	Privileged EXEC	`enable`
`Router(config)#`	Global configuration	`configure terminal`
`Router(config-if)#`	Interface	`interface fastEthernet 0/0`
`Router(config-router)#`	Routing protocol / sub-interface	`router ospf 1 / interface fa0/0.10`

> **WARNING** Router interfaces are shut down by default: every interface you use needs **no shutdown**. Save with **copy running-config startup-config** (or **write memory**) before closing the file.

A.5 Workflow for every experiment

Plan the addressing table, **build** with the right cables, **configure** end devices then routers/switches, **verify** with ping and show commands (capture the screenshots listed at the end of each experiment), and **save** the .pkt file.

EXPERIMENT 2

Subnetting

Divide 192.168.10.0/24 into two /25 subnets joined by a router

Aim

To subnet the Class C network 192.168.10.0/24 using a /25 mask, configure IP addresses on devices, and verify communication within and between the subnets through a router.

Devices required

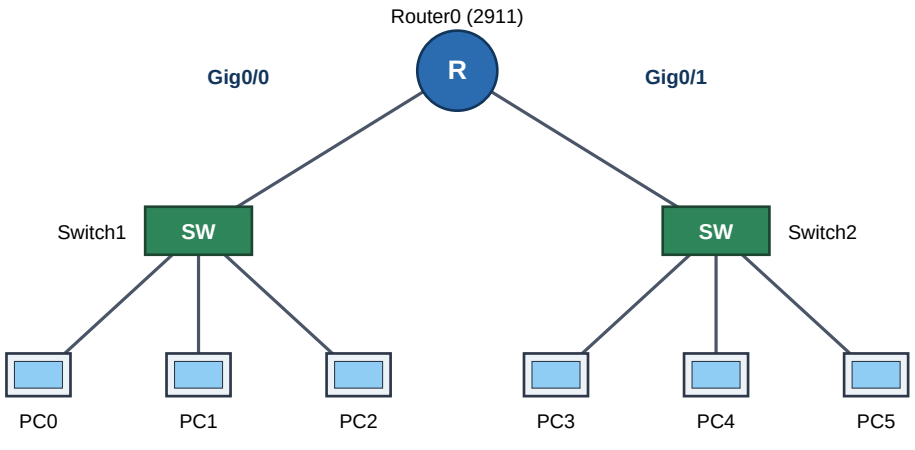
Device	Model	Qty
Router	2911	1
Switch	2960	2
PC	PC-PT	6
Cables	Copper straight-through, as required	-

Subnetting calculation

- Borrow **1 host bit**: 2^1 = **2 subnets**. New prefix = 24 + 1 = **/25**, mask **255.255.255.128**.
- Hosts per subnet = 2^7 - 2 = **126**. Block size = 256 - 128 = **128**, so subnets start at 0 and 128.

Parameter	Subnet 1	Subnet 2
Network address	192.168.10.0	192.168.10.128
First host	192.168.10.1	192.168.10.129
Last host	192.168.10.126	192.168.10.254
Broadcast address	192.168.10.127	192.168.10.255

Topology



Addressing table

Device	Interface	IP address	Sub net	Mask	Gateway
PC0	Fa0	192.168.10.2	#1	255.255.255.128	192.168.10.1
PC1	Fa0	192.168.10.3	#1	255.255.255.128	192.168.10.1
PC2	Fa0	192.168.10.126	#1	255.255.255.128	192.168.10.1
PC3	Fa0	192.168.10.130	#2	255.255.255.128	192.168.10.129
PC4	Fa0	192.168.10.131	#2	255.255.255.128	192.168.10.129
PC5	Fa0	192.168.10.254	#2	255.255.255.128	192.168.10.129
Router0	Gig0/0	192.168.10.1	#1	255.255.255.128	-
Router0	Gig0/1	192.168.10.129	#2	255.255.255.128	-

PC2 and PC5 use the **last usable host** of each subnet. That is deliberate: it proves your boundaries are right.

Procedure

1. Place 6 PCs, 2 switches (2960) and 1 router (2911). Connect PC0, PC1, PC2 to Switch1 and PC3, PC4, PC5 to Switch2 with straight-through cables.
2. Connect Switch1 to Router0 Gig0/0 and Switch2 to Router0 Gig0/1 (straight-through).
3. On each PC open **Desktop, IP Configuration, Static** and enter the address, mask 255.255.255.128 and the gateway of its own subnet.
4. Configure the router:

```
enable
configure terminal
hostname Router0
interface gigabitEthernet 0/0
 ip address 192.168.10.1 255.255.255.128
 no shutdown
interface gigabitEthernet 0/1
 ip address 192.168.10.129 255.255.255.128
 no shutdown
end
copy running-config startup-config
```

5. Switch to Simulation Mode, send a ping from a PC in Subnet #1 to a PC in Subnet #2 and watch the packet cross the router.

Verification

- `show ip interface brief` on the router: Gig0/0 and Gig0/1 both **up/up** with the correct addresses.
- `show ip route`: two **C** routes, 192.168.10.0/25 and 192.168.10.128/25.
- PC0 to PC1 (same subnet) and PC0 to PC4 (192.168.10.131, different subnet) must both succeed.

> **EXPECTED** The first ping often shows one timeout while ARP resolves. Ping again to get 4/4 replies.

Screenshots for the record

Subnet calculation, network topology, PC IP configuration, router configuration, show ip interface brief, ping results (same and different subnet), simulation mode.

Viva questions

- Why can the first and last address of each subnet not be assigned to hosts?
- How many subnets and hosts would a /26 give from the same network?
- Why does each subnet need its own default gateway?
- What is the difference between subnetting and supernetting?

EXPERIMENT 3

Simulation of LAN Topology with DNS and DHCP

Two LANs joined by a router, with DNS, web and DHCP servers

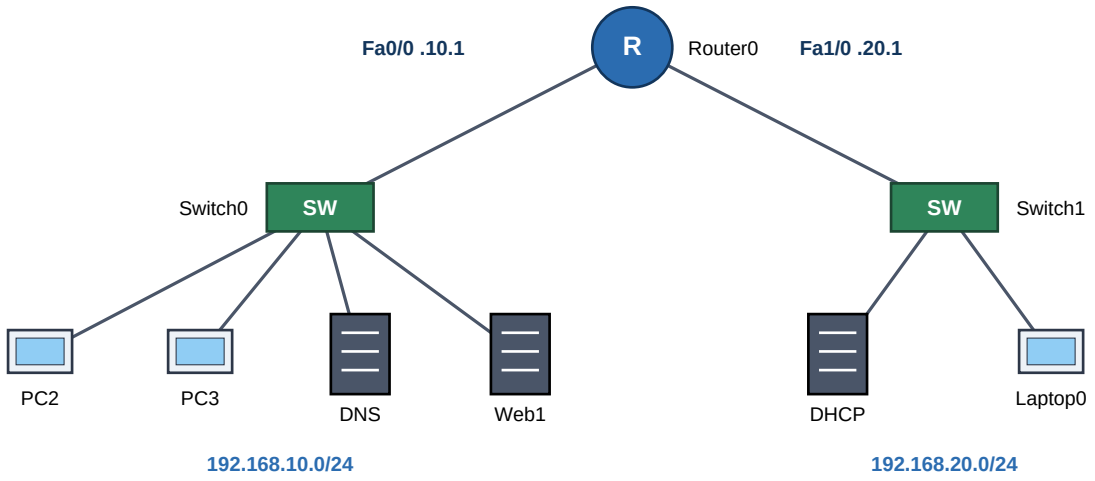
Aim

To design a LAN topology in Cisco Packet Tracer, configure DNS and DHCP services, investigate DNS and HTTP, and test connectivity using ICMP (ping).

Devices required

Device	Model	Qty
Router	Router-PT (Fa0/0, Fa1/0)	1
Switch	Switch0, Switch1	2
PC / Laptop	PC2, PC3, Laptop0	3
Servers	DNS Server, Web Server1, DHCP Server	3
Cables	Copper straight-through	-

Topology



Addressing table

Device	Interface	IP address	Mask	Gateway
Router0	Fa0/0	192.168.10.1	255.255.255.0	-
Router0	Fa1/0	192.168.20.1	255.255.255.0	-
DNS Server	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC2	NIC	192.168.10.3	255.255.255.0	192.168.10.1
Web Server1	NIC	192.168.10.4	255.255.255.0	192.168.10.1
PC3	NIC	192.168.10.5	255.255.255.0	192.168.10.1
DHCP Server	NIC	192.168.20.2	255.255.255.0	192.168.20.1
Laptop0	NIC	DHCP (from 192.168.20.10)	255.255.255.0	192.168.20.1 (from DHCP)

Procedure

1. Build the topology. Switch0 (DNS, Web, PC2, PC3) connects to Router0 Fa0/0; Switch1 (DHCP server, Laptop0) connects to Router0 Fa1/0. Use straight-through cables.
2. Switches need no configuration: all ports are up by default.
3. Configure the router so both LANs can talk:

```
enable
configure terminal
hostname Router0
interface fastEthernet 0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface fastEthernet 1/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
end
write memory
```

4. **DNS Server**: Desktop, IP Configuration: IP 192.168.10.2, mask 255.255.255.0, gateway 192.168.10.1. Services, DNS: turn **On**, add Name `www.myweb.com`, Type A Record, Address `192.168.10.4`, click **Add**.
5. **Web Server1**: IP 192.168.10.4, gateway 192.168.10.1. Services, HTTP: turn On (optionally edit the HTML page).
6. **DHCP Server**: IP 192.168.20.2, mask 255.255.255.0, gateway 192.168.20.1. Services, DHCP: turn **On** and fill the pool:

```
Pool Name       : Net2
Default Gateway : 192.168.20.1
DNS Server      : 192.168.10.2
Start IP Address: 192.168.20.10
Subnet Mask     : 255.255.255.0
Maximum Users   : 50
```

7. Click **Save**. Check that the pool appears in the table at the bottom of the DHCP page.
8. **PC2 and PC3**: static IPs from the table, gateway 192.168.10.1, DNS server 192.168.10.2. **Laptop0**: Desktop, IP Configuration, select **DHCP**.

> **NOTE** The DHCP server and Laptop0 are in the same LAN, so no DHCP relay is needed. The pool must hand out gateway 192.168.20.1 and DNS 192.168.10.2, otherwise Laptop0 can reach its own LAN but not 192.168.10.0/24.

Verification

- Laptop0 ipconfig: address from 192.168.20.10 upwards, gateway 192.168.20.1, DNS 192.168.10.2.
- PC3: `ping 192.168.10.4` (same LAN) and `ping www.myweb.com` (DNS in action).
- PC3 Web Browser: `http://www.myweb.com` opens the Cisco Packet Tracer page.
- Laptop0 ping 192.168.10.4 (another LAN through the router) should succeed. An unsuccessful ping to a non-existent host such as 192.168.10.100 shows "Request timed out".

> **TIP** If Laptop0 gets gateway 0.0.0.0 or DNS 192.168.20.2, the DHCP server is still answering from its default **serverPool**. Edit the pool named serverPool with the values above (or make sure your own pool is the only one) and click Save, then renew on Laptop0.

Screenshots for the record

Topology, PC IP configuration, DNS settings, web page access, DHCP settings, DHCP address on Laptop0, successful ping to a server, successful ping to another LAN, unsuccessful ping.

Viva questions

- Explain the DORA process of DHCP.
- Why is a different subnet for the DHCP server possible only with a relay agent?

- What does an A record store?
- Which transport protocol and port do DNS and HTTP use?

EXPERIMENT 4

Static Routing for Inter-Network Communication

Two LANs connected through a WAN (serial) link

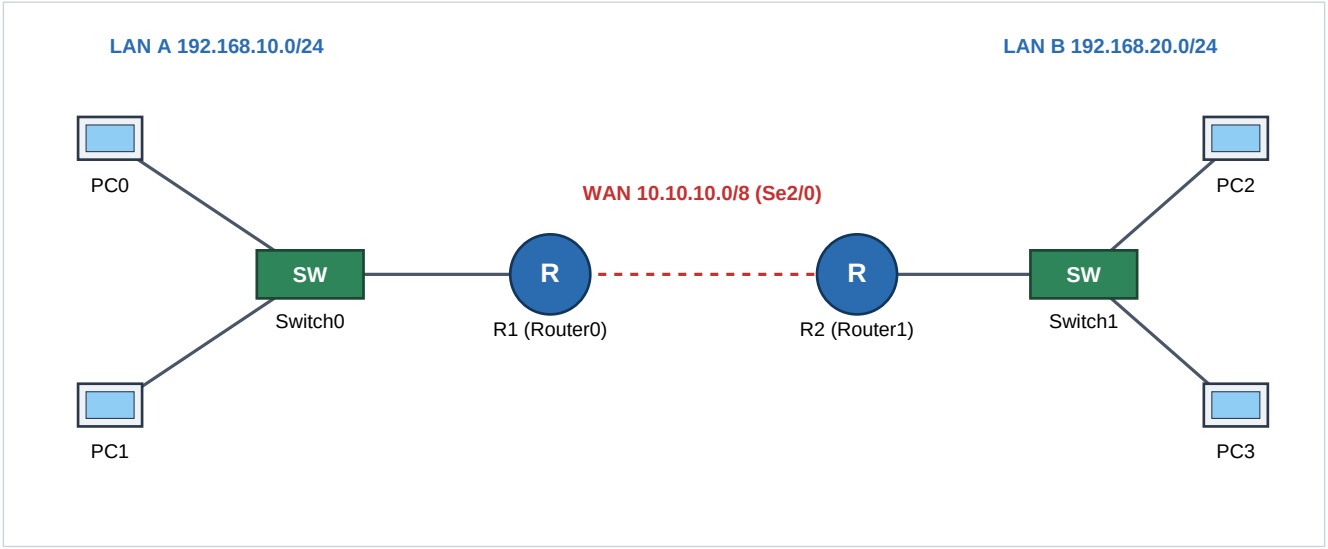
Aim

To design a topology of two LANs connected through a WAN link and establish communication between the different networks using static routing.

Devices required

Device	Model	Qty
Router	Router-PT (R1 = Router0, R2 = Router1)	2
Switch	2960 (S1, S2)	2
PC	PC-PT	4
Cables	Straight-through (PC-switch-router), Serial DCE (R1-R2)	-

Topology



Addressing table

Device	Interface	IP address	Subnet mask	Gateway
R1	Fa0/0	192.168.10.1	255.255.255.0	NA
R1	Serial2/0	10.10.10.1	255.0.0.0	NA
R2	Fa0/0	192.168.20.1	255.255.255.0	NA
R2	Serial2/0	10.10.10.2	255.0.0.0	NA
PC0	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC1	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC2	NIC	192.168.20.4	255.255.255.0	192.168.20.1
PC3	NIC	192.168.20.2	255.255.255.0	192.168.20.1

Procedure

1. Place 2 routers, 2 switches and 4 PCs. Use straight-through cables for PC to switch and switch to router (Fa0/0). Connect R1 Serial2/0 to R2 Serial2/0 with a **Serial DCE** cable.
2. Configure the PCs under Desktop, IP Configuration from the table. Before routes exist, PC0 cannot ping PC3.
3. **Router R1**: (the DCE end of the cable, marked with a clock icon, needs the clock rate)

```
enable
configure terminal
hostname R1
interface FastEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface Serial2/0
 ip address 10.10.10.1 255.0.0.0
 clock rate 64000
 no shutdown
exit
```

4. **Router R2**:

```
enable
configure terminal
hostname R2
interface Serial2/0
 ip address 10.10.10.2 255.0.0.0
 no shutdown
interface FastEthernet0/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
```

5. **Static routes**. R1 uses an exit interface, R2 uses a next-hop address; both forms are valid:

```
! ---- on R1 (to reach LAN B) ----
ip route 192.168.20.0 255.255.255.0 Serial2/0

! ---- on R2 (to reach LAN A) ----
ip route 192.168.10.0 255.255.255.0 10.10.10.1
```

6. Save both routers with `copy running-config startup-config`.

Verification

- `show ip route` on each router: **C** for connected networks, **S** for the static route.
- PC0 ping 192.168.20.2 (PC3) and PC3 ping 192.168.10.2 (PC0) must succeed.
- Simulation mode: follow the path PC, switch, R1, R2, switch, PC. The echo reply takes the reverse path.

```
R1# show ip route
C    10.0.0.0/8 is directly connected, Serial2/0
C    192.168.10.0/24 is directly connected, FastEthernet0/0
S    192.168.20.0/24 is directly connected, Serial2/0
```

> **TIP** Both routers need a route. If one direction is missing, ping shows "Destination host unreachable" or timeouts even though the other router is correct.

Screenshots for the record

Topology, PC configuration, R1 and R2 configuration, static route commands, show ip route (with S entry), ping PC0 to PC3 and PC3 to PC0, simulation packet flow.

Viva questions

- What is the administrative distance of a static route?
- Which side of a serial link needs the clock rate and why?
- Difference between next-hop and exit-interface static routes?
- Why is static routing not suitable for large networks?

EXPERIMENT 5

Dynamic Routing using OSPF

Three routers in a serial triangle learning routes automatically

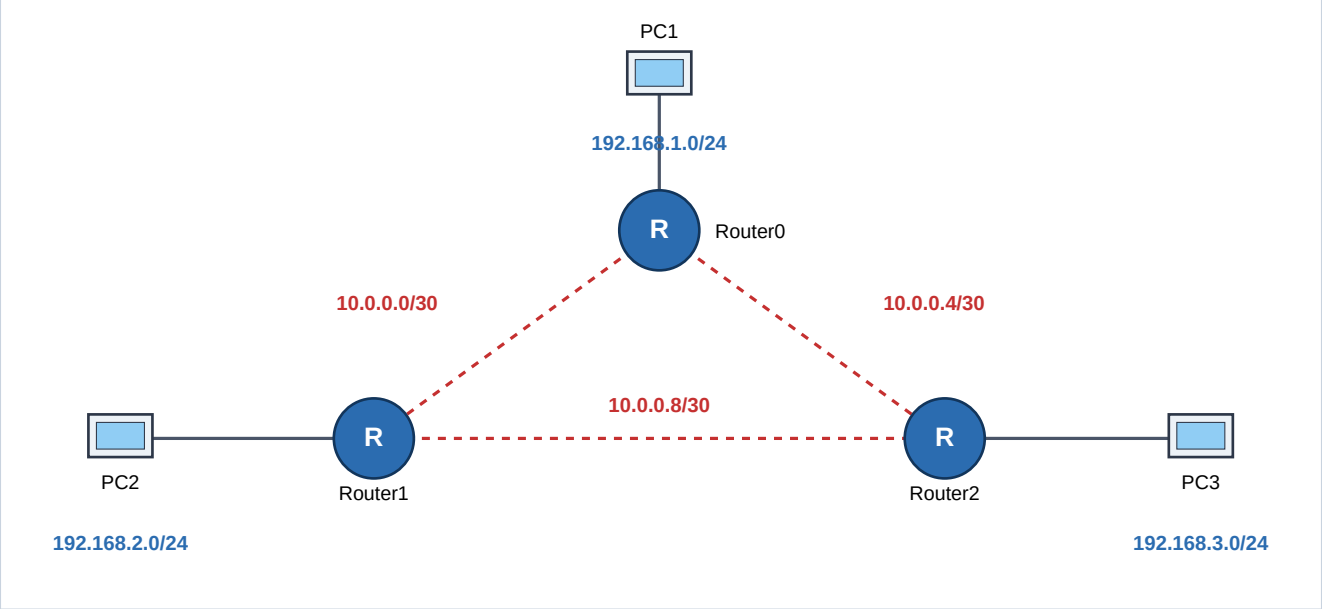
Aim

To configure single-area OSPF on three routers so that all LANs are reachable through dynamically learned routes, and to verify neighbours and routing tables.

Devices required

Device	Model	Qty
Router	Router-PT (Router0, Router1, Router2)	3
PC	PC1, PC2, PC3	3
Cables	Copper cross-over (PC-router), Serial DCE (router-router)	-

Topology



Addressing table

Device	Interface	IP address	Subnet mask	Link
PC1	NIC	192.168.1.2 (gw 192.168.1.1)	255.255.255.0	LAN 1
PC2	NIC	192.168.2.2 (gw 192.168.2.1)	255.255.255.0	LAN 2
PC3	NIC	192.168.3.2 (gw 192.168.3.1)	255.255.255.0	LAN 3
Router0	Fa0/0	192.168.1.1	255.255.255.0	LAN 1
Router0	Serial2/0	10.0.0.1	255.255.255.252	to Router1 (DCE)
Router0	Serial3/0	10.0.0.5	255.255.255.252	to Router2 (DCE)
Router1	Fa0/0	192.168.2.1	255.255.255.0	LAN 2
Router1	Serial2/0	10.0.0.2	255.255.255.252	to Router0
Router1	Serial3/0	10.0.0.9	255.255.255.252	to Router2 (DCE)
Router2	Fa0/0	192.168.3.1	255.255.255.0	LAN 3
Router2	Serial2/0	10.0.0.6	255.255.255.252	to Router0
Router2	Serial3/0	10.0.0.10	255.255.255.252	to Router1

Cabling: Router0 Se2/0 to Router1 Se2/0, Router0 Se3/0 to Router2 Se2/0, Router1 Se3/0 to Router2 Se3/0. Your record text uses Serial0/0 and 0/1 but its screenshots show Se2/0 and Se3/0; use whatever your router shows.

Theory in brief

- **OSPF** is a link-state protocol using Dijkstra SPF; administrative distance **110**; cost is based on bandwidth.
- **Network statements** use wildcard masks: /24 is 0.0.0.255, /30 is 0.0.0.3.
- **Router ID** uniquely identifies each router (set manually here as 1.1.1.1, 2.2.2.2, 3.3.3.3).

Procedure

1. Build the triangle and set PC addresses and gateways (Desktop, IP Configuration).
2. **Interface configuration.** Router0 example (Router1 and Router2 follow the table):

```
enable
configure terminal
hostname Router0
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
interface Serial2/0
 ip address 10.0.0.1 255.255.255.252
 clock rate 64000
 no shutdown
interface Serial3/0
 ip address 10.0.0.5 255.255.255.252
 clock rate 64000
 no shutdown
end
```

3. Clock rate goes on the DCE end only: Router0 Se2/0 and Se3/0, and Router1 Se3/0. Verify with `show ip interface brief` (all up/up).
4. **Enable OSPF** on each router:

```
! ---- Router0 ----
router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
 end
 wr

! ---- Router1 ----
router ospf 1
 router-id 2.2.2.2
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
 end
 wr

! ---- Router2 ----
router ospf 1
 router-id 3.3.3.3
 network 192.168.3.0 0.0.0.255 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
 end
 wr
```

5. Use the **same process ID on all routers** (the record text says 2 but its screenshots show ospf 1; either works as long as it is consistent). Wait about 30 seconds for the console message *ADJCHG ... from LOADING to FULL*.

Verification

- `show ip route`: networks learned from other routers carry the code **O** with **[110/cost]**.
- `show ip protocols`: shows the OSPF process, router ID and the networks being advertised. A router ID of 192.168.1.1 means the router-id command did not take effect (clear with `clear ip ospf process`).
- `show ip ospf neighbor`: each router lists two neighbours in state FULL.
- Ping **PC1 to 192.168.2.2 and 192.168.3.2** (use PC addresses, not .10).

```
Router0# show ip route
C    10.0.0.0/30 is directly connected, Serial2/0
C    10.0.0.4/30 is directly connected, Serial3/0
O    10.0.0.8/30 [110/128] via 10.0.0.2, Serial2/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
O    192.168.2.0/24 [110/65] via 10.0.0.2, Serial2/0
O    192.168.3.0/24 [110/65] via 10.0.0.6, Serial3/0
```

Failover observation (optional)

Shut down Router0 Serial2/0. After the dead timer the route to 192.168.2.0 moves to the path via Router2. Bring it back with no shutdown. This is the main advantage over static routing.

Screenshots for the record

Topology, PC configuration, interface configuration on each router, OSPF configuration, show ip interface brief, show ip route, show ip protocols, successful pings.

Viva questions

- Why does OSPF use wildcard masks?
- What is the role of area 0?
- Compare OSPF with static routing and with RIP.
- What are the default hello and dead timers on Ethernet and why do serial links behave differently?

EXPERIMENT 6

Intra-VLAN and Inter-VLAN Routing

Separate broadcast domains on one switch, then route between them

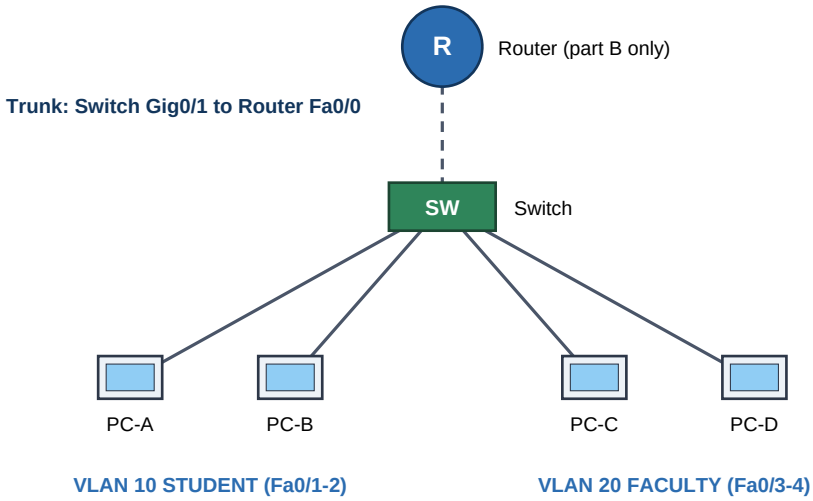
Aim

To configure intra-VLAN and inter-VLAN routing using VLANs, a Layer-2 switch and a router (router-on-a-stick), and to verify communication within and between VLANs.

Devices required

Device	Model	Qty
PC	PC-A, PC-B, PC-C, PC-D	4
Switch	2960	1
Router	Router-PT (part B)	1
Cables	Copper straight-through	-

Topology



Addressing table

Device	VLAN	Switch port	IP address	Mask	Gateway
PC-A	10	Fa0/1	192.168.1.2	255.255.255.0	192.168.1.1
PC-B	10	Fa0/2	192.168.1.3	255.255.255.0	192.168.1.1
PC-C	20	Fa0/3	172.16.1.2	255.255.255.0	172.16.1.1
PC-D	20	Fa0/4	172.16.1.3	255.255.255.0	172.16.1.1
Router Fa0/0.10	10	Trunk	192.168.1.1	255.255.255.0	-
Router Fa0/0.20	20	Trunk	172.16.1.1	255.255.255.0	-

Part A: Intra-VLAN

1. Place the 2960 switch and 4 PCs. Connect PC-A to Fa0/1, PC-B to Fa0/2, PC-C to Fa0/3 and PC-D to Fa0/4. Set the PC addresses and gateways.

2. Create the VLANs and assign access ports:

```
enable
configure terminal
vlan 10
 name STUDENT
vlan 20
 name FACULTY
exit
interface range fastEthernet 0/1 - 2
 switchport mode access
 switchport access vlan 10
interface range fastEthernet 0/3 - 4
 switchport mode access
 switchport access vlan 20
end
show vlan brief
```

3. VLAN names cannot contain spaces: use `STUDENT`, not "vlan student". Port numbers must match your switch (some 2950 models in your screenshots use Fa0/1, Fa1/1, Fa2/1...; use the ports you actually cabled).
4. Test: PC-A to PC-B (192.168.1.3) and PC-C to PC-D (172.16.1.3) **succeed**. PC-B to PC-C **fails** because the VLANs are separated and there is no router yet.

Part B: Inter-VLAN routing (router on a stick)

1. Add the router. Connect **Switch Gig0/1** to **Router Fa0/0** with a straight-through cable.
2. Make the switch port a trunk:

```
interface gigabitEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20
end
```

3. Configure sub-interfaces on the router. Do **not** put an IP address on the physical interface:

```
enable
configure terminal
interface fastEthernet 0/0
 no ip address
 no shutdown
interface fastEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.1 255.255.255.0
interface fastEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 172.16.1.1 255.255.255.0
end
copy running-config startup-config
```

> **WARNING** The "% 192.168.1.0 overlaps with FastEthernet0/0" error seen in the record appears when the physical interface already holds an address in the same network. Run **no ip address** on Fa0/0 first.

Verification

- `show vlan brief`: VLAN 10 (STUDENT) has Fa0/1-2; VLAN 20 (FACULTY) has Fa0/3-4.
- `show interfaces trunk`: Gi0/1 is trunking with VLANs 10 and 20 allowed.
- Router: `show ip interface brief` shows Fa0/0.10 and Fa0/0.20 up/up; `show ip route` has two connected networks.
- PC-A to PC-C and PC-B to PC-D now **succeed** (first ping may time out while ARP resolves).

Screenshots for the record

Intra-VLAN topology, show vlan brief, simulation pings (PC-A to PC-B success, PC-B to PC-C failed before routing), inter-VLAN topology, PC configuration, VLAN creation, router sub-interface configuration, show ip interface brief, successful inter-VLAN ping.

Viva questions

- What is a VLAN and how does it reduce broadcast traffic?
- Access port versus trunk port?
- Why is this method called router-on-a-stick and what is its limitation?
- What is the native VLAN?

EXPERIMENT 7

IPv6 Addressing using SLAAC

Hosts configure their own global IPv6 address from router advertisements

Aim

To design a topology and configure IPv6 using Stateless Address Autoconfiguration (SLAAC), and verify end-to-end connectivity between hosts.

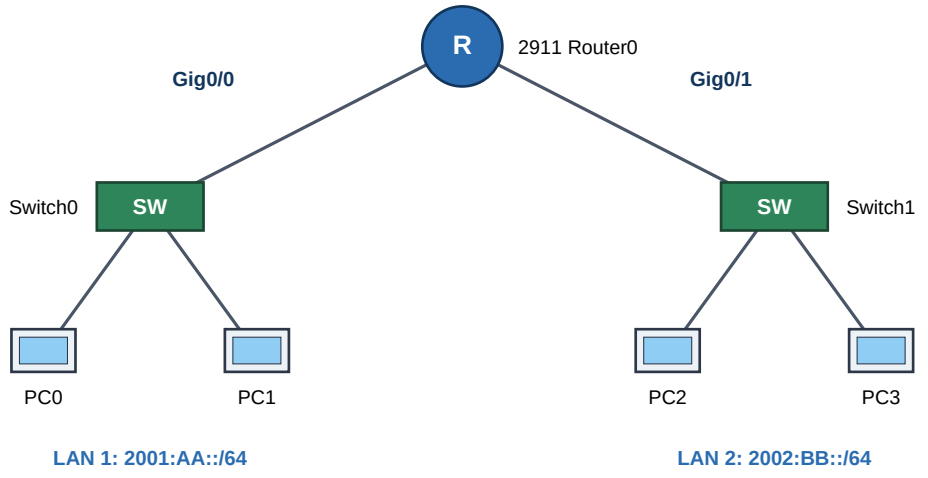
Theory in brief

- **SLAAC**: the router sends Router Advertisements (RA) with the /64 prefix; the host builds its own interface ID using **EUI-64** (MAC split, FFFE inserted, 7th bit flipped).
- **ipv6 unicast-routing** makes the router forward IPv6 and send RAs. **eui-64** on the router interface does the same address construction for the router itself.
- The host's default gateway is the router **link-local** address (FE80::...).

Devices required

Device	Model	Qty
Router	2911 (Gig0/0, Gig0/1)	1
Switch	2960-24TT	2
PC	PC0, PC1, PC2, PC3	4
Cables	Copper straight-through	-

Topology



Device	Interface	IPv6 address	Prefix	Default gateway
Router0	Gig0/0	2001:AA::/64 eui-64	/64	NA
Router0	Gig0/1	2002:BB::/64 eui-64	/64	NA
PC0, PC1	NIC	Automatic (SLAAC)	/64	Automatic
PC2, PC3	NIC	Automatic (SLAAC)	/64	Automatic

Procedure

1. Place 1 router (2911), 2 switches (2960-24TT) and 4 PCs. Connect PC0 and PC1 to Switch0, PC2 and PC3 to Switch1, then Switch0 to Router Gig0/0 and Switch1 to Router Gig0/1.
2. On each PC: Desktop, IP Configuration. Under **IPv6 Configuration** select **Automatic**. Leave IPv4 empty.
3. Configure the router:

```
enable
configure terminal
ipv6 unicast-routing
interface gigabitEthernet 0/0
 ipv6 address 2001:AA::/64 eui-64
 ipv6 enable
 no shutdown
interface gigabitEthernet 0/1
 ipv6 address 2002:BB::/64 eui-64
 ipv6 enable
 no shutdown
end
copy running-config startup-config
```

4. Wait a few seconds; each PC fills its IPv6 address, link-local address and gateway by itself.

> **NOTE** Both LANs hang off the same router, so no routing protocol is needed. The aim in the current record mentions RIPng, but it is not required here (it only matters with more than one router).

Verification

- PC: `ipconfig` shows a global address starting with 2001:AA: (LAN 1) or 2002:BB: (LAN 2), a FE80:: link-local address and the gateway FE80::....
- Compare the last 64 bits with the PC MAC address (Config tab) to confirm EUI-64.
- From a LAN 1 PC: `ping` the IPv6 address of a LAN 2 PC (copy it from that PC's ipconfig).
- Router: `show ipv6 interface brief`.

```
PC> ipconfig
   Link-local IPv6 Address.....: FE80::230:F2FF:FEB0:462D
   IPv6 Address................: 2001:AA::230:F2FF:FEB0:462D
   Default Gateway.............: FE80::20A:F3FF:FED2:7D01
```

> **EXPECTED** Your interface IDs depend on each MAC address and will differ from the sample.

Screenshots for the record

Network topology, router IPv6 configuration, ipconfig and ping from PC (first LAN), ipconfig on a PC in the second LAN, simulation view.

Viva questions

- Difference between SLAAC, stateless DHCPv6 and stateful DHCPv6?
- What is EUI-64? Derive the interface ID for a given MAC.
- What does ipv6 unicast-routing do?
- Why is NAT not needed in IPv6?

EXPERIMENT 8

Static NAT, Dynamic NAT and PAT

Translate private inside addresses at the edge router

Aim

To configure and verify Static NAT, Dynamic NAT and PAT on a router so internal private hosts can communicate with an external network while conserving public IP addresses.

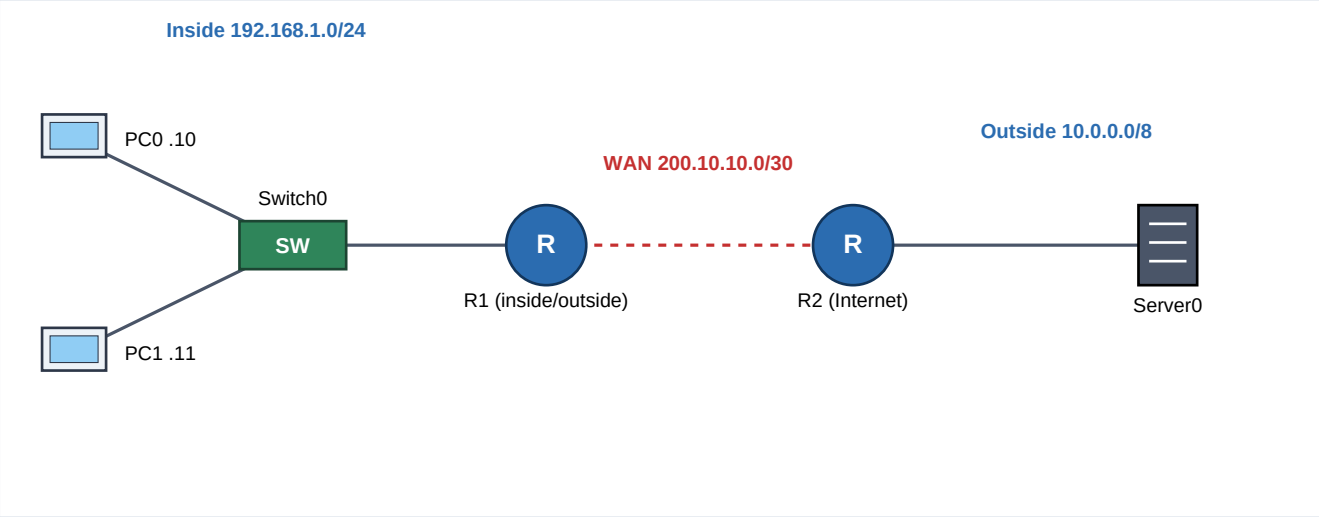
Theory in brief

Term	Meaning
Static NAT	Fixed one-to-one mapping (inside local to inside global).
Dynamic NAT	One-to-one mapping from a pool of public addresses; the pool can run out.
PAT (overload)	Many inside hosts share one public address, separated by port numbers.

Devices required

Device	Model	Qty
Routers	R1 edge (Router0), R2 Internet (Router1)	2
Switch	Switch0	1
PC	PC0, PC1	2
Server	Server0	1
Cables	Straight-through, serial DCE (R1-R2)	-

Topology



Addressing table

Device	Interface	IP address	Mask	Role
PC0	NIC	192.168.1.10	255.255.255.0	Gateway 192.168.1.1
PC1	NIC	192.168.1.11	255.255.255.0	Gateway 192.168.1.1
R1	Fa0/0	192.168.1.1	255.255.255.0	ip nat inside
R1	Serial2/0	200.10.10.1	255.255.255.252	ip nat outside (DCE)
R2	Serial2/0	200.10.10.2	255.255.255.252	ISP side
R2	Fa0/0	10.0.0.1	255.0.0.0	toward server
Server0	NIC	10.0.0.10	255.0.0.0	Gateway 10.0.0.1

Procedure: base configuration

1. Build the topology, set the PC and server addresses, enable the HTTP service on the server if you want a web test.
2. Configure R1 and R2:

```
! ---- R1 (edge router) ----
hostname R1
interface fastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 ip nat inside
 no shutdown
interface serial2/0
 ip address 200.10.10.1 255.255.255.252
 clock rate 64000
 ip nat outside
 no shutdown
exit
ip route 0.0.0.0 0.0.0.0 200.10.10.2

! ---- R2 (Internet router) ----
hostname R2
interface serial2/0
 ip address 200.10.10.2 255.255.255.252
 no shutdown
interface fastEthernet0/0
 ip address 10.0.0.1 255.0.0.0
 no shutdown
exit
! route to the PUBLIC addresses R1 translates to (not to 192.168.1.0)
ip route 200.10.10.8 255.255.255.248 200.10.10.1
```

3. Before NAT, ping 10.0.0.10 from PC0: it **fails**. The Internet router has no route to private 192.168.1.0/24, which is exactly the problem NAT solves.

> **NOTE** The record gives R2 a route to 192.168.1.0/24, which makes pings work even without NAT. For a proper demonstration, R2 should know only the public range (200.10.10.8/29) that R1 translates into.

Part A: Static NAT

```
! on R1: PC0 permanently appears as 200.10.10.10
ip nat inside source static 192.168.1.10 200.10.10.10
```

Ping 10.0.0.10 from PC0, then run `show ip nat translations` on R1: the static entry is listed. PC1 still cannot reach the server.

Part B: Dynamic NAT

```
! on R1: who may be translated, and the pool of public addresses
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat pool NAT_POOL 200.10.10.11 200.10.10.12 netmask 255.255.255.248
ip nat inside source list 1 pool NAT_POOL
```

Ping the server from PC1: it now gets 200.10.10.11. The ACL must match the **inside** network (192.168.1.0), not the Internet side.

Part C: PAT (overload)

```
! on R1: remove the pool and the static entry, then overload the outside interface
no ip nat inside source list 1 pool NAT_POOL
no ip nat inside source static 192.168.1.10 200.10.10.10
clear ip nat translation *
ip nat inside source list 1 interface serial2/0 overload
```

Ping the server from PC0 and PC1: both share 200.10.10.1 and differ only by port number.

Verification

```
R1# show ip nat translations
Pro  Inside global        Inside local       Outside local      Outside global
icmp 200.10.10.1:1        192.168.1.10:1     10.0.0.10:1        10.0.0.10:1
icmp 200.10.10.1:2        192.168.1.11:2     10.0.0.10:2        10.0.0.10:2
```

- `show ip nat statistics`: total translations, hits and misses, and which interfaces are inside and outside.
- `show running-config | include ip nat`, `show access-lists`, `show ip route`.
- Simulation mode: open the PDU before and after R1 and compare the source IP address.

> **TIP** If translations do not appear, check: **ip nat inside / outside** are on the right interfaces, the ACL matches 192.168.1.0, and R1 has the default route.

Screenshots for the record

Topology, R1 configuration, R2 configuration, ping from PC to router, show ip nat translations and statistics (one set per part: static, dynamic, PAT).

Viva questions

- Define inside local, inside global, outside local and outside global.
- When is static NAT preferred over dynamic NAT?
- How does PAT distinguish hosts sharing one public address?
- What problems does NAT cause for end-to-end protocols?

EXPERIMENT 9

Network Topologies: Bus, Star, Ring and Mesh

Build each topology with Ethernet switches only

Aim

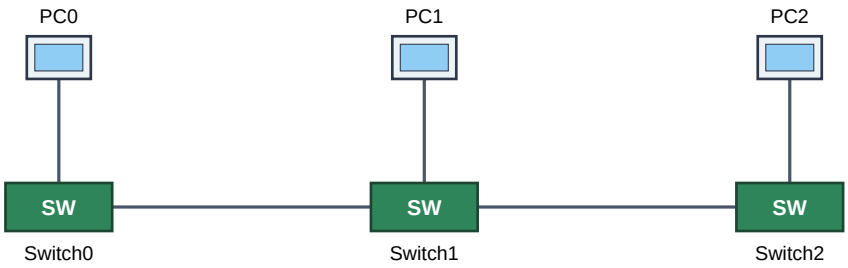
To build bus, star, ring and mesh topologies using Ethernet LAN cabling and 2960 switches only, assign static IP addresses to the PCs, and verify connectivity using ping.

Common setup

- All hosts are in one LAN **192.168.1.0/24**, mask 255.255.255.0, **no gateway** (no router).
- PC addresses: 192.168.1.10, .11, .12, .13 (one per PC in each topology). Switch management address on VLAN 1 (optional): 192.168.1.2.
- PC to switch: copper straight-through. Switch to switch: copper cross-over (or Automatic). Build each topology in its own file or separate area.
- Wait until link lights turn green. In ring and mesh topologies one or more ports stay **orange**: this is spanning tree blocking redundant links.

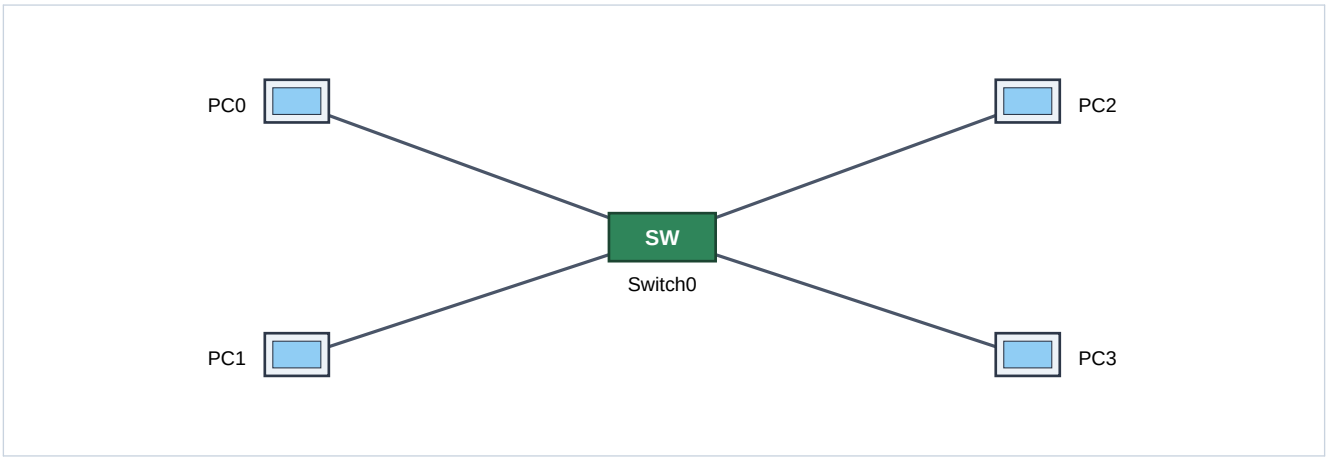
Topology	Switches	PCs	Description
Bus	3 (in a chain)	3	Switches linked in a line, one PC on each switch
Star	1	4	All PCs connected directly to one central switch
Ring	4	4	Four switches in a closed loop, one PC on each
Mesh	4	4	Every switch linked to every other (6 links), one PC on each

9.1 Bus topology



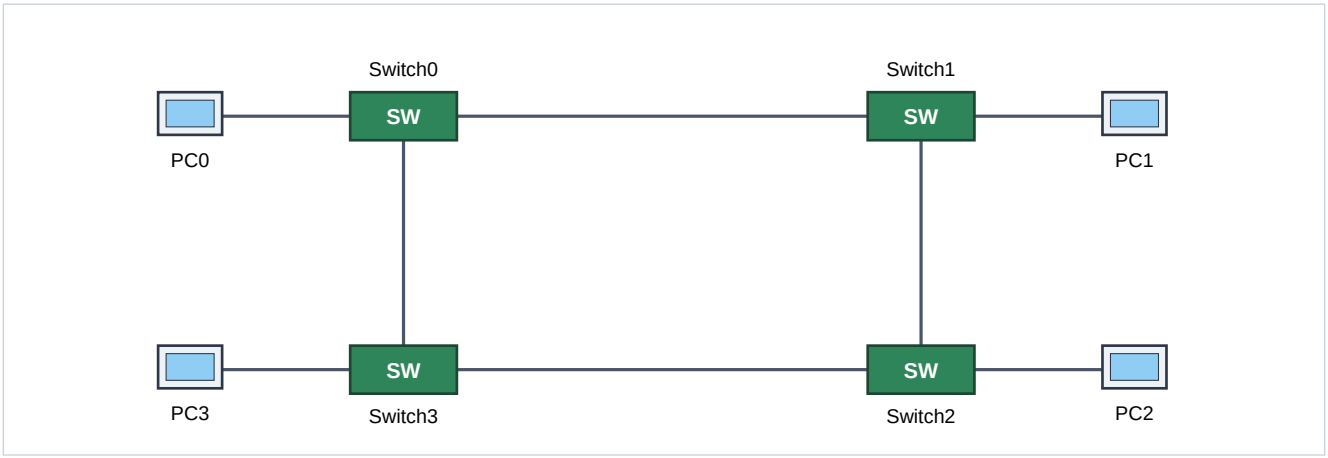
1. Place three switches in a row and one PC under each. Link Switch0-Switch1 and Switch1-Switch2 with cross-over cables.
2. Set PC IPs 192.168.1.10, .11, .12. Ping every PC from PC0 once the ports are green.

9.2 Star topology



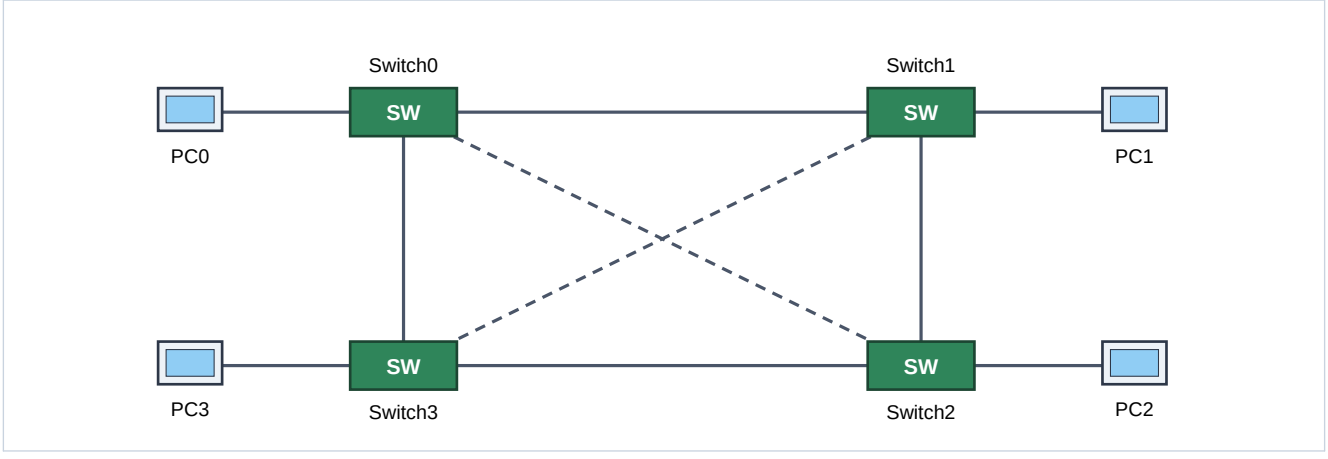
1. Place one switch and four PCs. Connect each PC to the switch with straight-through cables.
2. Set PC IPs 192.168.1.10 to .13 and ping from PC0 to all others. Check `show mac address-table` on the switch.

9.3 Ring topology



1. Place four switches and four PCs. Join the switches in a closed loop with cross-over cables and attach one PC to each switch.
2. Wait about 30 seconds: one inter-switch link shows an **orange** dot because spanning tree blocks it to prevent a loop.
3. Set IPs and ping PC0 to every PC. Run `show spanning-tree` to see the blocked port. Delete one active link and watch the blocked link recover.

9.4 Mesh topology



1. Start from the ring and add the two diagonal links (dashed). A full mesh of n nodes needs **n(n-1)/2** links: 4 x 3 / 2 = **6**.
2. Several ports will stay orange (blocked by spanning tree). Ping all PCs and confirm 100% success.

Switch configuration (any switch, as in the record)

```
enable
configure terminal
hostname S1
interface vlan 1
 ip address 192.168.1.2 255.255.255.0
 no shutdown
exit
line console 0
 password cisco
 login
exit
enable secret class
end
copy running-config startup-config
```

Use a different management address on each switch (.2, .3, .4, .5) if you configure more than one. Two devices with the same IP would cause an address conflict.

Verification

- PC0: `ping 192.168.1.11`, `.12`, `.13` must give 100% success (use the Add Simple PDU table if you prefer).
- `show ip interface brief`, `show vlan brief`, `show mac address-table` and `show running-config` on a switch.
- Ring and mesh: `show spanning-tree` shows the root bridge and the blocked ports.

> **NOTE** Packet Tracer has no coaxial bus or token ring media. The bus is modelled as a chain of switches; ring and mesh run Spanning Tree Protocol, so the physical topology is a ring or mesh but frames travel on a loop-free tree.

Screenshots for the record

Each topology (bus, star, ring, mesh), successful pings from PC0 to the other PCs, show running-config, show vlan brief and show mac address-table from a switch, PC IP configuration.

Topology	Links for n nodes	Fault tolerance	Cost
Bus	n (chain)	Poor: a break splits the network	Low
Star	n	Central switch is a single point of failure	Medium
Ring	n	One break survivable (STP recovery)	Medium
Mesh	n(n-1)/2	Excellent	High

Viva questions

- Why do ring and mesh need spanning tree?
- How many cables does a full mesh of 8 devices need?
- Which topology is used in modern Ethernet LANs and why?
- Difference between physical and logical topology?

Section B: Troubleshooting and reference

B.1 Troubleshooting checklist

Symptom	Likely cause	Fix
Link dot red or serial down	Wrong cable, interface shut, or missing clock rate	no shutdown; clock rate 64000 on the DCE end
First ping times out	ARP resolving	Ping again
Destination host unreachable	Missing route or wrong gateway	Check PC gateway; show ip route on every router
One-way ping works	Missing return route / NAT range not routed back	Add reverse route (Ex 4) or route to NAT pool (Ex 8)
Ping to wrong IP times out	Host does not exist (e.g. .10 instead of .2)	Read the addressing table
Laptop gets wrong gateway/DNS	serverPool answering instead of your pool	Edit serverPool or delete the extra pool; Save
OSPF neighbour missing	Wrong network/wildcard, serial down	show ip ospf neighbor; check network statements
Sub-interface "overlaps" error	IP on physical interface	no ip address on Fa0/0 first
VLAN hosts cannot reach gateway	Trunk missing or wrong dot1Q ID	show interfaces trunk; match dot1Q number to VLAN
NAT table empty	inside/outside swapped or ACL wrong	show ip nat statistics; ACL must match 192.168.1.0
Orange ports on switches	STP timers or blocked redundant link	Wait 30 s; normal in ring and mesh
Config lost on reopen	Not saved	copy running-config startup-config

B.2 Command cheat sheet

Command	Purpose
`show ip interface brief`	Interface status and IP (up/up)
`show running-config`	Full active configuration
`show ip route`	Routing table (C, S, O)
`show ip protocols`	Routing protocol details, router ID
`show ip ospf neighbor`	OSPF adjacencies
`show vlan brief`	VLANs and ports
`show interfaces trunk`	Trunk ports and allowed VLANs
`show ip nat translations / statistics`	NAT mappings and counters
`show ipv6 interface brief`	IPv6 addresses
`show spanning-tree`	Root bridge, blocked ports
`show mac address-table`	MAC addresses learned by a switch
`ping, tracert, ipconfig, nslookup`	PC tools

Section C: Fix these in the record before submission

While building this guide from the record I found several contradictions. They are easy to fix and a faculty member checking the record is likely to notice them. The guide already uses the corrected version.

Exp	What the record shows	Correction
3	Addressing table does not match the steps (PC0 .2 vs PC2 .3, DNS .3 vs .2, PC3 on 192.168.20.4, gateways swapped). Laptop0 screenshot shows gateway 0.0.0.0 and DNS 192.168.20.2, and its cross-LAN ping fails.	Use the step values (DNS .2, PC2 .3, Web .4, PC3 .5, DHCP .2 on 192.168.20.0). Make sure the Net2 pool, not serverPool, serves Laptop0, then retake the ping screenshot.
4	WAN written as 10.10.10.0/8. Screenshots 3 to 5 show R2 with dot1Q sub-interfaces and 172.16.1.1 (from the VLAN lab), a ping to 192.168.2.10 failing, and a routing table with no S route.	WAN network is 10.0.0.0/8. Retake screenshots 3 to 5 from this lab: R1/R2 serial config, show ip route with S entry, ping PC0 to PC3.
5	Title repeats Experiment 4 (static routing). Addressing table repeats 192.168.x.2 for every device. Text uses ospf 2, screenshot shows ospf 1 and router ID 192.168.1.1. Ping tests 192.168.2.10 (times out). Screenshot 7 is labelled show ip interface brief but shows show ip route.	Retitle as OSPF. Replace the table with the one in this guide (10.0.0.0/30, 0.4/30, 0.8/30). Use one process ID and set router-id. Ping 192.168.2.2 / 192.168.3.2. Add a real show ip interface brief screenshot.
6	VLAN name "vlan student" has a space. "int gig6/1" does not exist on the switch. IP address placed on Fa0/0 caused overlap errors. PC-C is .2 in the table and .3 in the text. Hardware line says 2950T Fa0/24 while topology uses Gig0/1.	Names STUDENT / FACULTY; trunk on gigabitEthernet 0/1 and router Fa0/0; no ip address on the physical interface; PC-C 172.16.1.2, PC-D 172.16.1.3.
7	Aim says RIPng is implemented but no RIPng commands are shown; device table says 2 routers while the topology has 1.	Remove RIPng from the aim (or add the commands) and list 1 router.
8	Dynamic NAT ACL permits 10.0.0.0/24 (the server side) and the pool is 203.0.113.x, unrelated to the 200.10.10.0 addressing. R2 has a route to 192.168.1.0, so NAT is not actually needed for the ping. NAT statistics show Hits 0.	ACL permits 192.168.1.0 0.0.0.255; pool 200.10.10.11-12; R2 routes to 200.10.10.8/29 via R1. Retake translations/statistics after real traffic.
9	Table says bus = 1 switch and mesh = 3 switches, but screenshots show 3 chained switches and 4 switches. Addressing table lists 4 PCs for all topologies. Screenshots 9 and 10 show a Router0 config with ACLs (from Experiment 10).	Update the table to bus 3 / mesh 4 and retake switch CLI screenshots (show running-config, show vlan brief, show mac address-table).